Results: Reproducibility Assessment

Results: Reproducibility Assessment I

An optional, **LLM-gated** stage decomposes each paper's described pipeline into a workflow graph of source, method, experiment, and sink steps, rates how reproducible each step is from the paper's own text, and combines a content score with a structural graph-coverage score into one composite reproducibility score per paper (geometric mean, so a paper cannot buy a high score by being strong on one axis alone). Across 0 scored papers the mean composite score is pending, with 0 papers falling below the configured low-score threshold.

Table 8 lists the papers with the lowest composite reproducibility scores, alongside their content and structural component scores.

Table 8. Low-scoring papers by composite reproducibility score.

Low-Scoring Papers I

Paper	Composite	Content	Structural

Gating and Defaults I

This stage is optional and gated by full-text availability. With no fulltext available and no language model configured, the stage is skipped and the reproducibility aggregates read *pending* — the same graceful-degradation convention used by the knowledge-graph assertion-extraction stage (see `02d_methods_knowledge_graph.md`). When fulltext is available and a language model is explicitly configured, the mean score, low-score count, and per-paper table are populated from extracted workflow graphs. The default fixture run does not imply that this optional model-backed assessment has occurred.

Interpretation I

A low composite score can reflect either weak content (the paper's own text does not describe its sources, methods, experiments, or outputs in enough detail to rate highly) or weak structure (the described steps do not chain into a coherent source-to-sink pipeline, or reference steps that were never themselves described). The two axes are reported separately in Table 8 precisely so a low composite score can be diagnosed rather than treated as a single undifferentiated verdict.